Solve the System:

$$\begin{cases} 2^x + 2^y = 10 \\ 2^{x+y} = 16 \end{cases} \Rightarrow 2^{x+y} = 2^x \cdot 2^y$$

$$\Rightarrow 2^x \cdot 2^y = 16$$

Lets say: $u = 2^x$, $v = 2^y$

$$\begin{cases} u + v = 10 \\ u \cdot v = 16 \end{cases} \Rightarrow v = 10 - u$$

$$\left.\begin{aligned} u(10-u) &= 16 \\ 10u - u^2 - 16 &= 0 \\ u^2 - 10u + 16 &= 0 \end{aligned}\right\} \begin{aligned} &u = 8 \ , \ u = 2 \\ &\Rightarrow 2^x = 8 \ , \ 2^x = 2 \end{aligned}$$

when $u = 8$:

$8v = 16$

$v = 2$

$\Rightarrow 2^y = 2$

$y = 1$

when $u = 2$

$2v = 16$

$v = 8$

$\Rightarrow 2^y = 8$

$y = 3$

also : $x = 1$, $x = 3$

$$S : \{(1, 3)(3, 1)\}$$